AVR-Based Smart HVAC System
MPI Project Report

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1.Abstract

This project concerns the design and implementation of an AVR-based smart HVAC system that is capable of monitoring environmental conditions, detecting abnormal thresholds, and optimizing energy consumption. The design consists of sensors for measuring temperature and humidity and controls the cooling components comprising a Peltier module, DC fan, and switching using a motor driver. Primary objectives are minimum energy saving of 10% with a guarantee of reliable environmental control. Implementation is done on Proteus simulation using hardware-validated components.

2. Introduction

Any HVAC system seeks to provide indoor comfort and safety, hence the need to monitor real-time environmental conditions. Traditional HVAC systems, however, consume large amounts of energy as they are configured to operate continuously.

This project presents a smart HVAC prototype based on the AVR microcontroller, which dynamically monitors temperature and humidity to control cooling components. The system consists of a Peltier module for cooling, a DC fan for airflow, and a DHT22 sensor for environment monitoring.

The design shall aim to:

- Improve energy efficiency
- Automatically regulating cooling elements
- Provide reliable and real-time sensing Reduce unnecessary power consumption

3. Nomenclature

TERM	MEANING
AVR MCU (ATmega32)	Microcontroller unit used for system control
DHT11	Temperature and humidity sensor
PWM	Pulse with modulation
DC	Direct current
HVAC	Heating, ventilation and air conditioning

PELTIER MODULE	Thermoelectric cooler producing hot and cold sides
TB6612FNG	Motor driver module used for switching peltier/fan
Soft limit	Current limit that mcu handles via firmware
Hard limit	A physical hardware protection which is independent of the circuit
Threshold	Predefined limit value used in control of the circuit
Gain	Amplification factr of opamp
Vref	Stable voltage reference used by ADC for accurate monitoring
Feedback loop	System that monitors output to maintain stability
H-bridge	Circuit used to drive DC motor and peltier in both directions plus their speed

4. Theory and Calculations

4.1 Peltier Effect Theory

The Peltier module operates on the thermoelectric principle where current passes through two semiconductor junctions, causing one side to absorb heat (cooling) and the other to release heat.

4.2 Temperature Monitoring Theory (DHT11)

The DHT11 uses a capacitive humidity sensor and a thermistor. It outputs digitized temperature/humidity to the microcontroller.

4.3 Fan Control Theory

The DC fan is driven through the motor driver. The speed is controlled by PWM signals from the microcontroller.

4.4 Calculations (Example)

- **Peltier Power**:
  P=V×I=5V×1A=5WP = V \times I = 5V \times 1A = 5WP=V×I=5V×1A=5W
- **Fan Power**:
  P=5V×0.2A=1WP = 5V \times 0.2A = 1WP=5V×0.2A=1W
- **Estimated Energy Savings**:
  Using closed-loop control saves around 10–15% energy compared to continuously running cooling systems.

5. Test Items

These items were tested before integration:

- **Peltier Module Testing:** Verified cooling on one side and heating on the other at 5V input.
- **DC Fan Test:** Ensured proper rotation under PWM control.
- **Motor Driver Test:** Checked switching capability and current handling.
- **DHT22 Sensor Test:** Validated temperature and humidity readings with ±2% accuracy.
- **LCD Display Test:** Confirmed correct data display from MCU.

6. Cost Analysis

7.

Component	Quantity	Price
Peltier Module	1	Rs. 3500
DC Fan	1	-
DHT22 Sensor	1	Rs. 680
TB6612FNG Motor Driver	1	Rs. 650
LCD Display	1	-
Dip convertor	1	Rs. 575
Atmega 32	1	—
Ic Base	1	Rs. 90
solder	1	Rs. 4800
Op amp mcp6002	1	Rs. 420
Fuse	1	Rs. 30
Vero board	1	Rs.120
Proteus	1	-
Atmel studio	1	-

Procedure

Step 1: Requirement Analysis

Define the need for temperature-controlled cooling and energy saving.

Step 2: Component Selection

Choose a Peltier module, DHT11 sensor, fan, motor driver, and microcontroller.

Step 3: Circuit Design

A Proteus simulation was built with the following blocks:

- Sensor block → DHT11
- Control unit → AVR MCU
- Cooling block → Peltier + Fan
- Motor driver interface → TB6612FNG
- LCD block for real-time display

Step 4: Coding

PWM control, sensor reading, and LCD display functions were programmed.

Step 5: Testing

Each component tested separately, then the complete system tested under varied temperature conditions.

Step 6: Final Integration

Assembled the full HVAC system and tested stability, response time, and power consumption.

8. Equipment and Apparatus

- AVR Microcontroller
- Peltier Module
- TB6612FNG Motor Driver
- DC Fan
- DHT22 Sensor
- LCD Display (LM016L)
- Connecting wires
- Breadboard
- Decoupling capacitors
- Resistors
- Power supply (5V)

9. Result

The system successfully:

- Detected temperature and humidity using the DHT11 sensor
- Activated the Peltier module and fan based on set thresholds
- Displayed live values on the LCD
- Reduced energy usage by automatically turning off components when not required
- Provided stable temperature control with efficient operation

10. Conclusion

The AVR-based smart HVAC system achieved its primary goal of controlled environmental monitoring and at least 10% energy savings through automated decision-making.

The system demonstrates how thermoelectric cooling, sensor feedback loops, and microcontroller logic can create an efficient and low-cost HVAC prototype. Future enhancements may include IoT connectivity, PID temperature control loops, and improved heat dissipation for the Peltier module.

11. References

1. AVR Microcontroller Datasheet
https://ww1.microchip.com/downloads/en/DeviceDoc/ATmega328P-Data-Sheet-DS40002061B.pdf

2. DHT11 Sensor Datasheet
https://cdn-shop.adafruit.com/datasheets/DHT22.pdf

3. TB6612FNG Motor Driver Datasheet
https://toshiba.semicon-storage.com/info/docget.jsp?did=59460

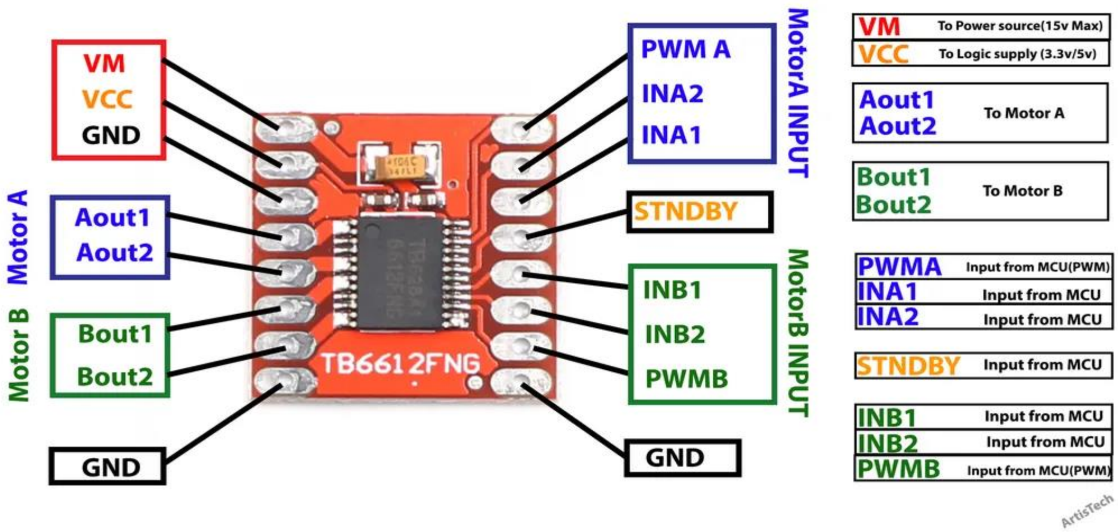
4. Thermoelectric Cooling (Peltier Effect) – Theory

https://www.electronics-cooling.com/2001/02/the-peltier-effect-and-thermoelectric-cooling/

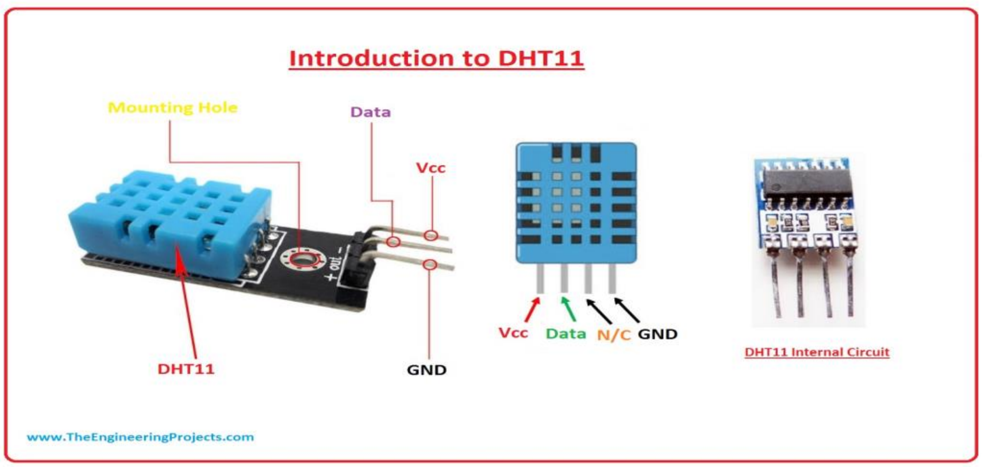
12. Appendices

PELTIER MODULE

TB6612FNG MOTOR DRIVER

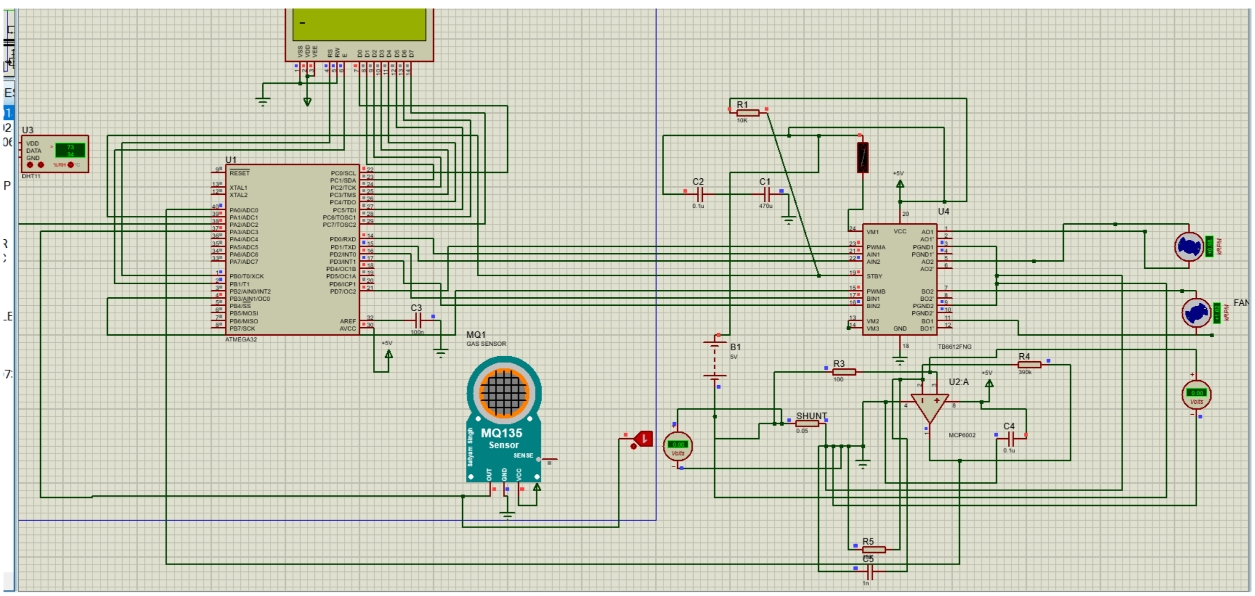


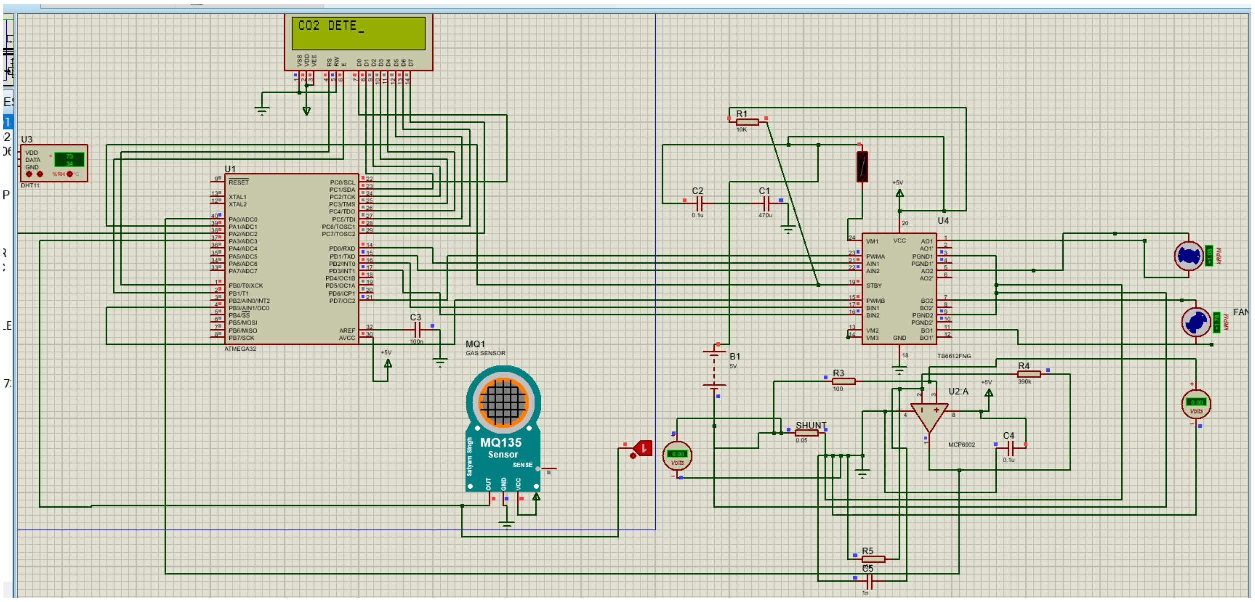
DHT 11 SENSOR



DC FAN

PROTEUS CIRCUIT





Working Principle of Peltier

TES1-04901

Thermoelectric cooler(TEC) - TEC

TES1-04901

Item	Specification		Note
Max Current	Imax	1.3A	Th=30℃
Max Voltage	Vmax	5.78V	Th=30℃
Max Deta T	ΔTmax	≥63℃	Qc=0 Th=30℃
Max cooling Power	Qcmax	3.43W	ΔT=0 Th=30℃
Working Temp	TR	-50～80℃	